Bloch Sphere and State Representation

Why Do We Need State Representation

Saying that a qubit is in superposition raises a natural question: where exactly is that qubit? In classical computing the answer is simple, since a bit is either 0 or 1. In quantum computing a qubit can exist in infinite possible combinations of |0> and |1>, so we need a way to describe how much of |0> exists, how much of |1> exists, and the relationship between them. This description is called quantum state representation.

Analogy: A Compass

A compass needle can point north, south, east, west, or any direction in between. Similarly, a qubit can exist in |0>, |1>, or any valid quantum combination between them. Instead of representing only two possibilities, we represent the qubit as a point in a state space.

1. State Representation

A quantum state describes everything we know about a qubit before measurement. Mathematically, it is written as |psi> = alpha|0> + beta|1>, where alpha is the amplitude of |0> and beta is the amplitude of |1>.

Important Rule

The probabilities must add up to 1, written as |alpha|^2 + |beta|^2 = 1. This is because after measurement the qubit must become either |0> or |1>, and there is no third outcome.

Three Key Examples

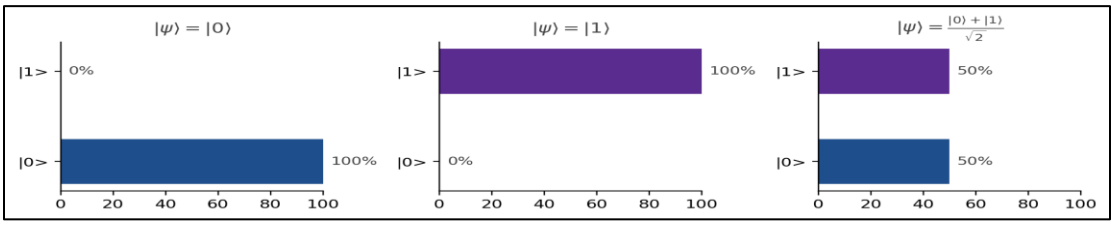
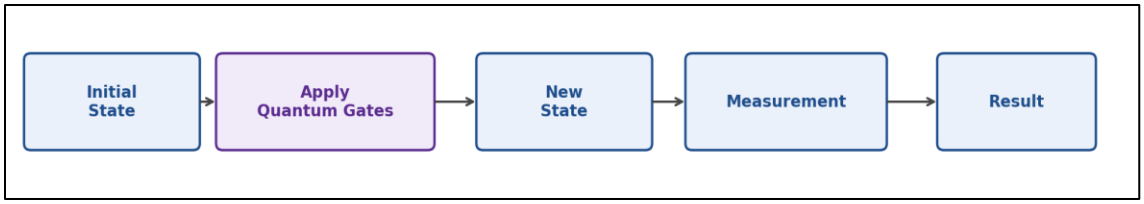


Figure 1: |0> and |1> are certain outcomes; equal superposition splits the probability evenly.

When |psi> = |0>, the qubit has a 100 percent chance of measuring 0. When |psi> = |1>, it has a 100 percent chance of measuring 1. When |psi> = (|0> + |1>) / square root of 2, there is a 50 percent chance of either outcome. This third case is one of the most important quantum states and is called equal superposition.

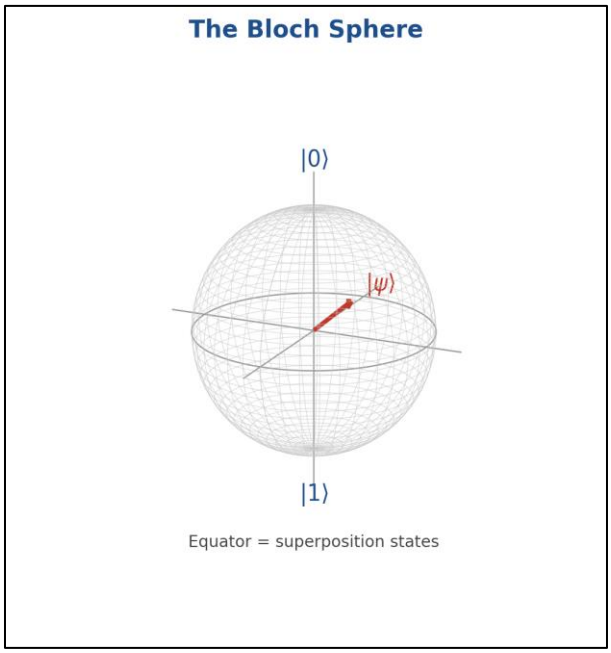
Why State Representation Matters

Quantum algorithms do not directly manipulate bits, they manipulate quantum states. Understanding state representation means understanding what the computer is actually processing.



2. The Bloch Sphere

The equation |psi> = alpha|0> + beta|1> is mathematically correct but difficult to visualise. Humans understand pictures better than equations, so scientists created a visual model called the Bloch Sphere. It is a 3D visualisation of a single qubit, where every valid pure qubit state corresponds to one point on the surface of the sphere, similar to using coordinates on a map.



Reading the Sphere

- North Pole represents |0>, a 100 percent probability of measuring 0
- South Pole represents |1>, a 100 percent probability of measuring 1
- Every other point represents a superposition state, containing information about both outcomes at once

Basis States

The Z-basis states, |0> and |1>, are the computational basis states we finally measure. The X-basis states, |+> and |->, are defined as (|0>+|1>)/root2 and (|0>-|1>)/root2, and are heavily used in quantum algorithms. The Y-basis states, |+i> and |-i>, introduce phase information, which is one of the key differences between classical and quantum computing.

How Quantum Gates Affect the Bloch Sphere

A qubit can be thought of as an arrow starting from the centre of the sphere. Quantum gates rotate this arrow to a new position, moving the qubit from its initial state to a new state. This is why quantum gates are described as rotations on the Bloch Sphere.

Limitation of the Bloch Sphere

The Bloch Sphere only works for a single qubit. When multiple qubits become entangled, the combined system can no longer be fully represented by one Bloch Sphere, and more advanced representations are needed.

3. Quick Comparison

Concept	Purpose
State representation	Mathematical description of a qubit
State vector	Equation describing the qubit
Bloch Sphere	Visual representation of the qubit
Quantum gates	Rotations on the Bloch Sphere
Measurement	Collapse to a single definite outcome

Summary: What to Remember

- A quantum state describes everything we know about a qubit before measurement
- |psi> = alpha|0> + beta|1>, where the probabilities |alpha|^2 and |beta|^2 must add up to 1
- Equal superposition gives a 50 percent chance of measuring 0 or 1
- The Bloch Sphere is a 3D visual model of a single qubit's state
- The poles represent |0> and |1>; every other point represents a superposition
- Z, X, and Y basis states describe different reference directions on the sphere
- Quantum gates rotate the qubit's position on the Bloch Sphere
- The Bloch Sphere only works for a single qubit, not entangled systems